E-Commerce Order Management System

Business Analysis Case Study

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Business Analyst Portfolio

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Project Overview

1. Project Title

E-Commerce Order Management System (OMS)

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2. Project Description

A centralized Order Management System designed to automate and streamline the complete order lifecycle, including customer order placement, payment processing, inventory validation, warehouse fulfillment, shipment tracking, and order delivery. The system aims to improve operational efficiency, reduce manual effort, and enhance customer satisfaction.

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3. Project Domain

E-Commerce / Retail

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4. Project Type

Business Analysis Case Study

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5. Methodology

Agile Scrum

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6. Project Purpose

To design an efficient and scalable Order Management System that simplifies order processing, improves inventory management, and provides customers with a seamless shopping experience.

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7. Project Vision

To create an integrated order management solution that enables faster order processing, accurate inventory tracking, and real-time visibility across the entire order lifecycle.

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8. Project Scope

In Scope

- User Registration and Login
- Product Browsing and Search
- Shopping Cart Management
- Checkout Process
- Payment Processing
- Order Management
- Inventory Validation
- Warehouse Processing
- Shipment Tracking
- Order Cancellation
- Returns and Refunds
- Admin Dashboard
- Reporting and Analytics

Out of Scope

- Supplier Management
- Manufacturing Operations
- Customer Loyalty Program
- International Tax Calculation
- Third-Party Marketplace Management

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9. Project Objectives

- Automate the order management process.
- Reduce order processing time.
- Improve inventory accuracy.
- Minimize manual intervention.
- Enhance customer experience.
- Provide real-time order tracking.
- Improve operational efficiency.

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10. Key Modules

- User Management
- Product Catalog
- Shopping Cart
- Checkout
- Payment Gateway
- Order Management
- Inventory Management
- Warehouse Management
- Shipping and Delivery
- Returns Management

- Notifications
- Reporting Dashboard
- Admin Portal

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11. Expected Deliverables

- Business Case
- Stakeholder Analysis
- AS-IS Process
- TO-BE Process
- Requirements Elicitation
- Business Requirements Document (BRD)
- Functional Requirements Document (FRD)
- User Stories
- Acceptance Criteria
- Process Diagrams
- Wireframes
- Data Analysis
- Dashboard
- Test Scenarios
- Business Benefits

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12. Expected Outcome

The proposed Order Management System will streamline business operations by automating order processing, improving inventory visibility, reducing operational delays, and delivering a better customer experience through an integrated and efficient platform.

Business Case

1. Business Problem

The existing order management process relies on multiple disconnected systems and manual coordination between departments such as sales, inventory, warehouse, and logistics. This results in delayed order processing, inventory mismatches, payment issues, limited order visibility, and increased customer complaints. The lack of a centralized system negatively impacts operational efficiency and customer satisfaction.

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2. Current Business Situation

The organization is experiencing continuous growth in online sales, leading to a higher volume of customer orders. However, the current process is unable to efficiently manage this increasing demand due to manual workflows and fragmented systems.

As a result, the business faces operational delays, inaccurate inventory information, order cancellations, and increased support requests from customers seeking order status updates.

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3. Business Need

The organization requires a centralized Order Management System that integrates order processing, payment validation, inventory management, warehouse operations, and shipment tracking into a single platform.

The new system should automate manual activities, improve collaboration between departments, and provide customers with real-time order updates.

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4. Opportunity Statement

Implementing an integrated Order Management System provides an opportunity to improve operational efficiency, reduce processing errors, enhance customer satisfaction, and support future business growth through automation and standardized business processes.

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5. Cost of Inaction

If the current process continues without improvement, the organization may experience:

- Increased order processing delays
- Higher order cancellation rates
- Inventory inconsistencies
- Growing operational costs

- Poor customer experience
- Loss of repeat customers
- Reduced competitive advantage
- Increased manual workload for employees

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6. Proposed Solution

Develop and implement a centralized E-Commerce Order Management System that automates the complete order lifecycle, including customer order placement, payment processing, inventory validation, warehouse fulfillment, shipment tracking, returns management, and reporting.

The proposed solution will provide real-time visibility across all business functions and improve coordination between departments.

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7. Expected Business Value

The proposed solution is expected to deliver the following benefits:

- Faster order processing
- Improved inventory accuracy
- Reduced manual effort
- Better order tracking capabilities
- Enhanced customer satisfaction
- Lower operational costs
- Improved employee productivity
- Better business reporting and decision-making

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8. Business Objectives

- Automate end-to-end order processing.
- Reduce order processing time.
- Improve inventory visibility.
- Increase order accuracy.
- Enhance customer experience.
- Minimize operational errors.
- Improve process efficiency.
- Support business scalability.

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9. Success Criteria

The project will be considered successful if it achieves the following outcomes:

- Order processing time reduced by at least 50%.
- Inventory accuracy exceeds 99%.

• Order cancellation due to stock issues is minimized.
• Customer satisfaction improves significantly.
• Manual processing effort is reduced.
• Real-time order tracking is available to customers and administrators.

10. Business Risks

Risk	Impact	Mitigation Strategy
Payment gateway failure	High	Use backup payment services
Inventory synchronization issues	High	Implement real-time inventory updates
System downtime	High	Deploy high-availability infrastructure
User resistance to change	Medium	Conduct user training sessions
Integration challenges	Medium	Perform thorough integration testing

11. High-Level Recommendation

Based on the current business challenges and future growth requirements, implementing a centralized E-Commerce Order Management System is recommended. The proposed solution will improve operational efficiency, reduce business risks, enhance customer satisfaction, and provide a scalable platform to support the organization's long-term digital transformation goals.

12. Conclusion

The implementation of an E-Commerce Order Management System represents a strategic investment that addresses existing operational inefficiencies while enabling sustainable business growth. By automating critical business processes and improving visibility across the order lifecycle, the organization can deliver a superior customer experience and achieve greater operational excellence.

Stakeholder Analysis

1. Introduction

Stakeholder Analysis is the process of identifying all individuals, groups, and departments that are involved in or affected by the E-Commerce Order Management System project. Understanding stakeholder roles, interests, and influence helps ensure effective communication, requirement gathering, and successful project delivery.

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2. Purpose of Stakeholder Analysis

The purpose of stakeholder analysis is to identify key stakeholders, understand their expectations and responsibilities, manage their involvement throughout the project lifecycle, and ensure that business objectives are achieved through effective collaboration.

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3. Stakeholder Identification

The following stakeholders have been identified for the E-Commerce Order Management System project:

- Project Sponsor
- Business Analyst
- Project Manager
- Product Owner
- Customers
- System Administrator
- Warehouse Team
- Inventory Team
- Finance Team
- Customer Support Team
- Delivery Partner
- Development Team
- Quality Assurance (QA) Team

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4. Stakeholder Roles and Responsibilities

Stakeholder	Role	Responsibility
Project Sponsor	Executive Decision Maker	Provides funding and approves the project
Business Analyst	Requirement Analyst	Gathers and documents business requirements
Project Manager	Project Lead	Manages project planning and execution

Stakeholder	Role	Responsibility
Product Owner	Business Representative	Prioritizes business requirements and backlog
Customer	End User	Places orders and uses the system
System Administrator	System Manager	Manages users and system configuration
Warehouse Team	Operations Team	Picks, packs, and dispatches orders
Inventory Team	Stock Manager	Maintains product inventory
Finance Team	Payment Management	Validates payments and processes refunds
Customer Support Team	Customer Service	Resolves customer issues and inquiries
Delivery Partner	Logistics	Delivers orders to customers
Development Team	Technical Team	Develops the application
QA Team	Testing Team	Validates system functionality and quality

5. Stakeholder Interest and Influence Matrix

Stakeholder	Interest	Influence	Management Strategy
Project Sponsor	High	High	Manage Closely
Product Owner	High	High	Manage Closely
Project Manager	High	High	Manage Closely
Business Analyst	High	High	Manage Closely
Customer	High	Medium	Keep Informed
Warehouse Team	High	Medium	Keep Informed
Inventory Team	High	Medium	Keep Informed
Finance Team	Medium	Medium	Keep Satisfied
Customer Support Team	Medium	Low	Keep Informed
Delivery Partner	Medium	Low	Monitor
Development Team	High	High	Manage Closely
QA Team	High	Medium	Keep Informed

6. Stakeholder Expectations

• Fast and accurate order processing.

- Real-time inventory updates.
- Secure payment processing.
- Easy order tracking.
- Reduced operational delays.
- Improved customer satisfaction.
- Reliable reporting and analytics.
- User-friendly system interface.

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7. Communication Plan

Stakeholder	Communication Method	Frequency
Project Sponsor	Project Status Report	Monthly
Project Manager	Team Meetings	Weekly
Product Owner	Sprint Review	End of Sprint
Business Analyst	Requirement Workshops	As Required
Development Team	Daily Stand-up Meeting	Daily
QA Team	Test Review Meeting	Weekly
Warehouse Team	Operational Updates	Weekly
Customer Support Team	Email / Meetings	Weekly

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8. Stakeholder Engagement Strategy

- Conduct requirement gathering workshops with business stakeholders.
- Organize regular sprint review meetings.
- Share project progress reports with management.
- Collect user feedback during User Acceptance Testing (UAT).
- Maintain continuous communication with development and testing teams.
- Address stakeholder concerns promptly through scheduled meetings.

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9. Stakeholder Challenges

- Conflicting business requirements.
- Resistance to process changes.
- Communication gaps between departments.
- Delayed feedback from stakeholders.
- Changing business priorities during project execution.

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10. Risk Mitigation

- Conduct regular stakeholder meetings.
- Maintain clear project documentation.
- Define stakeholder responsibilities early.
- Use requirement traceability to manage changes.
- Establish an effective change management process.

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11. Success Factors

Effective stakeholder engagement will contribute to:

- Better requirement quality.
- Faster decision-making.
- Reduced project risks.
- Improved collaboration.
- Higher user adoption.
- Successful project implementation.

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12. Conclusion

Stakeholder Analysis plays a critical role in the successful delivery of the E-Commerce Order Management System project. Identifying stakeholders, understanding their expectations, and maintaining effective communication throughout the project lifecycle will ensure alignment between business needs and technical solutions, ultimately contributing to the successful achievement of project objectives.

AS-IS Process Analysis

1. Introduction

The AS-IS Process Analysis documents the current order management process followed by the organization before implementing the proposed E-Commerce Order Management System. It provides a clear understanding of the existing workflow, identifies operational inefficiencies, and highlights areas that require improvement.

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2. Purpose

The purpose of the AS-IS Process Analysis is to examine the current business process, identify bottlenecks and challenges, and establish a baseline for designing an improved TO-BE process.

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3. Current Business Process

The existing order management process follows these steps:

1. Customer visits the e-commerce website.
2. Customer searches for products.
3. Customer adds products to the shopping cart.
4. Customer proceeds to checkout.
5. Customer enters shipping details.
6. Customer completes payment.
7. Order details are sent to the sales team.
8. Inventory team manually verifies product availability.
9. Warehouse team receives the order request.
10. Warehouse staff picks and packs the products.
11. Delivery partner is assigned manually.
12. Shipment is dispatched.
13. Customer receives the order.
14. Customer contacts support for any issues or returns.

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4. Current Process Flow

Customer

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Browse Products

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Add to Cart

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Checkout

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Payment

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Manual Inventory Verification

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Warehouse Processing

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Manual Delivery Assignment

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Order Dispatch

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Delivery to Customer

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5. Existing Challenges

The current process faces several operational challenges:

- Manual inventory verification causes delays.
- Order processing depends on multiple disconnected systems.
- Limited visibility into order status.
- Manual coordination between departments.
- Delayed shipment assignment.
- Increased risk of human error.
- Higher order cancellation due to stock mismatches.
- Slow return and refund processing.

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6. Pain Points

Customer Pain Points

- Delayed order confirmation.

- Lack of real-time order tracking.
- Slow customer support response.
- Order cancellations after payment.
- Delayed refunds.

Business Pain Points

- Manual operational effort.
- Inventory inaccuracies.
- Duplicate data entry.
- Increased operational costs.
- Difficulty generating reports.
- Poor interdepartmental communication.

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7. Root Cause Analysis

The primary causes of the existing issues include:

- Lack of an integrated order management platform.
- Manual inventory validation process.
- Absence of automated workflow management.
- Limited system integration between departments.
- Dependence on manual communication.
- No centralized reporting mechanism.

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8. Process Inefficiencies

- Repeated manual verification activities.
- Delays in warehouse notification.
- Manual shipment allocation.
- Slow exception handling.
- Inconsistent order status updates.
- Redundant data processing.

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9. Business Impact

The current process negatively impacts business performance by:

- Increasing operational costs.
- Reducing employee productivity.
- Lowering customer satisfaction.
- Increasing order processing time.
- Increasing cancellation rates.
- Affecting customer retention and brand reputation.

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10. Opportunities for Improvement

The organization can improve its operations by:

- Automating order processing.
- Integrating inventory management with order placement.
- Implementing real-time inventory updates.
- Automating shipment assignment.
- Providing live order tracking.
- Centralizing reporting and analytics.
- Reducing manual intervention across departments.

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11. Key Findings

The AS-IS analysis indicates that the existing order management process is heavily dependent on manual activities and disconnected systems. These inefficiencies lead to delays, errors, and poor customer experience, highlighting the need for a centralized and automated Order Management System.

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12. Conclusion

The current order management process is no longer sufficient to support the growing needs of the business. Implementing an integrated E-Commerce Order Management System will address the identified pain points, streamline operations, improve collaboration across departments, and enhance the overall customer experience.

TO-BE Process Analysis

1. Introduction

The TO-BE Process Analysis describes the future state of the order management process after implementing the proposed E-Commerce Order Management System. The redesigned process focuses on automation, system integration, improved efficiency, and enhanced customer experience.

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2. Purpose

The purpose of the TO-BE Process Analysis is to define an optimized business process that eliminates manual activities, reduces operational delays, improves accuracy, and supports the organization's future growth objectives.

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3. Proposed Business Process

The proposed order management process follows these steps:

1. Customer logs into the e-commerce platform.
2. Customer searches and selects products.
3. Customer adds products to the shopping cart.
4. System validates product availability in real time.
5. Customer proceeds to checkout.
6. Customer enters shipping information.
7. Customer completes secure online payment.
8. System automatically generates the order.
9. Inventory is updated automatically.
10. Warehouse team receives an automated fulfillment request.
11. Warehouse staff picks, packs, and prepares the order.
12. System assigns a delivery partner automatically.
13. Customer receives order confirmation and tracking details.
14. Customer tracks the shipment in real time.
15. Order is delivered successfully.
16. Customer can initiate returns or refunds through the system if required.

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4. Future Process Flow

Customer

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Login

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Browse Products

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Add to Cart

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Real-Time Inventory Validation

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Checkout

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Online Payment

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Automatic Order Creation

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Automatic Inventory Update

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Warehouse Processing

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Automatic Delivery Assignment

↓

Shipment Tracking

↓

Order Delivery

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Returns and Refund Processing (if applicable)

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5. Process Improvements

The proposed solution introduces the following improvements:

- Automated order creation.
- Real-time inventory validation.
- Automatic inventory updates.
- Automated warehouse notifications.
- Automatic delivery partner assignment.
- Real-time shipment tracking.
- Digital return and refund process.
- Centralized reporting and monitoring.

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6. Business Benefits of the TO-BE Process

Customer Benefits

- Faster order confirmation.
- Real-time order tracking.
- Improved shopping experience.
- Faster refunds and returns.
- Better communication through automated notifications.

Business Benefits

- Reduced manual effort.
- Improved inventory accuracy.
- Faster order fulfillment.
- Lower operational costs.
- Better coordination across departments.
- Increased employee productivity.
- Improved customer retention.

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7. Automation Opportunities

The proposed system automates several business activities, including:

- Inventory validation.
- Order creation.
- Inventory updates.
- Warehouse notifications.
- Delivery partner assignment.
- Order status notifications.
- Customer email and SMS alerts.
- Management reporting.

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8. Future State Capabilities

The new Order Management System will provide:

- Centralized order management.
- Integrated inventory management.
- Secure payment processing.
- Real-time shipment tracking.
- Automated notifications.
- Dashboard and analytics.
- Role-based access control.
- Scalable architecture for future expansion.

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9. Comparison Between AS-IS and TO-BE Process

AS-IS Process	TO-BE Process
Manual inventory verification	Real-time inventory validation
Manual order processing	Automated order processing
Separate departmental systems	Integrated centralized system
Manual shipment assignment	Automatic shipment assignment
Limited order visibility	Real-time order tracking
Manual reporting	Automated dashboards and reports
High operational effort	Streamlined automated workflows
Higher processing time	Faster order fulfillment

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10. Expected Performance Improvements

- Reduced order processing time.
- Improved inventory accuracy.
- Reduced order cancellations.
- Faster warehouse operations.
- Increased customer satisfaction.
- Improved operational efficiency.
- Better reporting and decision-making.
- Enhanced scalability for business growth.

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11. Success Factors

The success of the TO-BE process depends on:

- Effective system integration.
- Accurate business requirements.
- User training and adoption.
- Reliable payment gateway integration.
- Real-time inventory synchronization.
- Continuous monitoring and process improvement.

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12. Conclusion

The TO-BE Process Analysis presents a future-state business process that leverages automation and system integration to improve operational efficiency and customer experience. By replacing manual workflows with a centralized Order Management System, the organization can achieve faster processing, greater accuracy, reduced operational costs, and long-term business scalability.

Requirements Elicitation

1. Introduction

Requirements Elicitation is the process of identifying, gathering, and understanding the business needs and stakeholder expectations for the E-Commerce Order Management System. This phase ensures that the proposed solution addresses business problems and aligns with organizational objectives.

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2. Purpose

The purpose of Requirements Elicitation is to collect accurate and complete business requirements from stakeholders, analyze their needs, and establish a clear foundation for designing and developing the proposed Order Management System.

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3. Objectives

- Understand the existing business process.
- Identify business challenges and pain points.
- Gather functional and non-functional requirements.
- Define stakeholder expectations.
- Document business rules and constraints.
- Ensure alignment between business goals and system capabilities.

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4. Stakeholders Involved

The following stakeholders participated in the requirements elicitation process:

- Project Sponsor
- Product Owner
- Business Analyst
- Project Manager
- Customers
- Warehouse Team
- Inventory Team
- Finance Team
- Customer Support Team
- Delivery Partner
- Development Team
- Quality Assurance Team

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5. Requirements Elicitation Techniques

The following techniques were used to gather requirements:

Interviews

One-on-one discussions were conducted with key stakeholders to understand business objectives, operational challenges, and system expectations.

Workshops

Collaborative sessions were organized with multiple stakeholders to discuss business processes, identify issues, and define future requirements.

Observation

Existing order processing activities were observed to understand current workflows and identify manual tasks and inefficiencies.

Document Analysis

Current business documents, order reports, inventory records, and operational procedures were reviewed to understand the existing system.

Brainstorming Sessions

Cross-functional teams participated in brainstorming sessions to generate ideas for improving the order management process.

Questionnaires and Surveys

Feedback was collected from customers and operational teams to identify user expectations and improvement opportunities.

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6. Business Requirements Identified

The following business requirements were identified during the elicitation process:

- Automate the order processing workflow.
- Provide real-time inventory validation.
- Enable secure online payment processing.
- Generate automatic order confirmations.
- Improve warehouse coordination.
- Provide shipment tracking functionality.
- Support order cancellation and returns.
- Generate business reports and dashboards.
- Improve customer communication through automated notifications.

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7. Functional Requirements Identified

The system should:

- Allow user registration and login.
- Display product catalog.
- Support product search and filtering.
- Enable shopping cart functionality.
- Process online payments.
- Generate unique order IDs.
- Update inventory automatically.
- Assign delivery partners automatically.
- Track order status.
- Manage returns and refunds.
- Generate reports for administrators.

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8. Non-Functional Requirements Identified

The system should:

- Respond within acceptable performance limits.
- Maintain high availability and reliability.
- Ensure secure payment processing.
- Protect customer data.
- Support scalability for future growth.
- Provide a user-friendly interface.
- Maintain data integrity.
- Support concurrent users without performance degradation.

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9. Business Rules Identified

- Orders can only be placed for available products.
- Payment must be successful before order confirmation.
- Inventory must be updated immediately after order placement.
- Each order must have a unique Order ID.
- Customers can cancel orders before shipment.
- Refunds should be processed only after return approval.
- Only authorized administrators can manage inventory.

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10. Assumptions

- Customers have internet access.
- Payment gateway services are available.
- Inventory data is maintained accurately.
- Warehouse operations support digital processing.
- Delivery partners are integrated with the system.

11. Constraints

- Limited project budget.
- Fixed implementation timeline.
- Integration with existing business systems.
- Compliance with security and privacy standards.
- Dependency on third-party payment gateway availability.

12. Key Findings

The elicitation process revealed that the existing order management process relies heavily on manual activities and disconnected systems. Stakeholders emphasized the need for automation, centralized data management, real-time inventory visibility, and improved customer communication to support business growth and operational efficiency.

13. Outcome of Requirements Elicitation

The requirements gathered during this phase provide the foundation for preparing the Business Requirements Document (BRD), Functional Requirements Document (FRD), user stories, acceptance criteria, and system design artifacts. These requirements will guide the development team in delivering a solution that meets business objectives and stakeholder expectations.

14. Conclusion

The Requirements Elicitation phase successfully identified the business needs, stakeholder expectations, and system requirements for the E-Commerce Order Management System. The information collected will serve as the basis for designing an efficient, scalable, and user-centric solution that improves operational performance and enhances the overall customer experience.

Business Requirements Document (BRD)

1. Introduction

The Business Requirements Document (BRD) defines the business needs and objectives for the E-Commerce Order Management System. It serves as a formal document that captures stakeholder expectations and outlines the business requirements that the proposed solution must fulfill.

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2. Purpose

The purpose of this document is to provide a clear understanding of the business requirements, project objectives, scope, and expected outcomes to ensure alignment among stakeholders and support successful project implementation.

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3. Business Objective

The primary objective of the project is to implement an integrated Order Management System that automates the complete order lifecycle, improves operational efficiency, enhances customer satisfaction, and supports business growth through streamlined processes.

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4. Business Problem Statement

The current order management process relies heavily on manual activities and disconnected systems, leading to delayed order processing, inventory inconsistencies, payment issues, operational inefficiencies, and poor customer experience.

The organization requires a centralized system to improve process efficiency and reduce operational risks.

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5. Project Scope

In Scope

- User Registration and Login
- Product Catalog Management
- Product Search and Filtering
- Shopping Cart Management
- Checkout Process
- Payment Processing
- Order Creation
- Inventory Validation
- Warehouse Processing
- Shipment Tracking

- Order Cancellation
- Returns and Refund Management
- Admin Dashboard
- Reporting and Analytics

Out of Scope

- Supplier Management
- Manufacturing Process
- Customer Loyalty Program
- International Tax Management
- Third-Party Marketplace Management

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6. Business Requirements

The proposed system shall:

- Automate order processing activities.
- Validate inventory availability in real time.
- Process secure online payments.
- Generate unique order confirmations.
- Update inventory automatically after order placement.
- Enable real-time shipment tracking.
- Support order cancellation and returns.
- Generate business reports and dashboards.
- Improve communication between departments.
- Enhance customer experience through automated notifications.

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7. Business Goals

- Improve operational efficiency.
- Reduce manual effort.
- Increase order processing speed.
- Improve inventory accuracy.
- Reduce order cancellations.
- Enhance customer satisfaction.
- Improve reporting capabilities.
- Support business scalability.

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8. Stakeholders

The key stakeholders involved in the project include:

- Project Sponsor
- Product Owner
- Business Analyst

- Project Manager
- Development Team
- Quality Assurance Team
- Customers
- Warehouse Team
- Inventory Team
- Finance Team
- Customer Support Team
- Delivery Partners

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9. Business Assumptions

- Customers will access the platform through the internet.
- Payment gateway services will be available.
- Inventory data will be maintained accurately.
- Warehouse operations will support digital processing.
- Delivery partners will integrate with the system.

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10. Business Constraints

- Fixed project budget.
- Defined implementation timeline.
- Existing system integration requirements.
- Compliance with security and privacy regulations.
- Dependence on third-party payment providers.

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11. Business Risks

- Payment gateway downtime.
- Inventory synchronization failures.
- Integration challenges.
- User resistance to system changes.
- High transaction volumes during peak sales periods.

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12. Critical Success Factors

The project will be considered successful if it achieves the following:

- Faster order processing.
- Accurate inventory management.
- Reduced operational costs.
- Improved customer satisfaction.
- Increased process automation.
- Better business reporting.
- Higher employee productivity.

13. Business Requirements Summary

Requirement ID	Business Requirement
BR-001	The system shall automate order processing.
BR-002	The system shall validate inventory in real time.
BR-003	The system shall support secure online payments.
BR-004	The system shall generate unique Order IDs.
BR-005	The system shall update inventory automatically.
BR-006	The system shall provide shipment tracking.
BR-007	The system shall support returns and refunds.
BR-008	The system shall generate management reports.
BR-009	The system shall send automated customer notifications.
BR-010	The system shall provide an administrative dashboard.

14. Expected Business Benefits

- Improved operational efficiency.
- Faster order fulfillment.
- Reduced manual processing.
- Better inventory control.
- Enhanced customer experience.
- Lower operational costs.
- Improved decision-making through reporting.
- Increased business scalability.

15. Approval Criteria

The Business Requirements Document will be considered approved when all identified stakeholders review and agree that the documented business requirements accurately represent the business needs and project objectives.

16. Conclusion

This Business Requirements Document establishes the foundation for the successful implementation of the E-Commerce Order Management System. It clearly defines the business needs, project scope, objectives, and expected outcomes, ensuring that all stakeholders have a shared understanding of the solution before proceeding to detailed functional design and development.

Functional Requirements Document (FRD)

1. Introduction

The Functional Requirements Document (FRD) defines the functional behavior of the E-Commerce Order Management System. It describes the system features, business functions, validation rules, inputs, outputs, and user interactions required to fulfill the business requirements documented in the BRD.

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2. Purpose

The purpose of this document is to provide detailed functional requirements that guide the design, development, testing, and implementation of the Order Management System.

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3. Scope

The FRD covers all functional modules related to customer order processing, inventory management, payment processing, warehouse operations, shipment tracking, returns management, and administrative reporting.

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4. Functional Modules

- User Management
- Product Catalog
- Shopping Cart
- Checkout
- Payment Processing
- Order Management
- Inventory Management
- Warehouse Management
- Shipment Tracking
- Returns and Refund Management
- Notifications
- Admin Dashboard

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5. Functional Requirements

FR-001: User Registration

Description: The system shall allow new users to create an account.

Input:

- Full Name
- Email Address
- Mobile Number
- Password

Output:

- User account created successfully.

Validation Rules:

- Email must be unique.
- Mobile number must be valid.
- Password must meet security standards.

FR-002: User Login

Description: The system shall authenticate registered users.

Input:

- Email Address
- Password

Output:

- Successful login and access to the user dashboard.

Validation Rules:

- Credentials must be valid.
- Invalid login attempts should display an error message.

FR-003: Product Search

Description: The system shall allow users to search and filter products.

Input:

- Product Name
- Category
- Price Range

Output:

- Matching products displayed.

FR-004: Shopping Cart

Description: The system shall allow users to add, update, and remove products from the shopping cart.

Input:

- Product Selection
- Quantity

Output:

- Updated shopping cart summary.

FR-005: Checkout

Description: The system shall allow customers to complete the purchase process.

Input:

- Shipping Address
- Payment Method

Output:

- Order summary before payment.

FR-006: Payment Processing

Description: The system shall process secure online payments.

Input:

- Payment Details

Output:

- Payment confirmation or failure notification.

Validation Rules:

- Payment must be authorized before order confirmation.

FR-007: Order Creation

Description: The system shall automatically generate an order after successful payment.

Output:

- Unique Order ID
- Order Confirmation

FR-008: Inventory Management

Description: The system shall validate product availability and update inventory automatically after order placement.

Output:

- Updated stock quantity.

FR-009: Warehouse Processing

Description: The system shall notify the warehouse team automatically for order fulfillment.

Output:

- Warehouse task created.

FR-010: Shipment Tracking

Description: The system shall provide real-time shipment tracking information to customers.

Output:

- Current shipment status.
- Estimated delivery date.

FR-011: Returns and Refunds

Description: The system shall allow customers to request product returns and refunds.

Output:

- Return request created.
- Refund status updated.

FR-012: Notifications

Description: The system shall send automated notifications for important order events.

Notification Events:

- Order Confirmation
- Payment Confirmation
- Shipment Dispatched
- Order Delivered
- Return Approved
- Refund Processed

FR-013: Admin Dashboard

Description: The system shall provide administrators with a dashboard to monitor business operations.

Dashboard Features:

- Total Orders
- Total Revenue
- Pending Orders
- Cancelled Orders
- Inventory Status
- Customer Statistics

---

6. Business Rules

- Products must be available before checkout.
- Payment must be completed before order confirmation.
- Inventory shall update automatically after successful order placement.
- Order IDs shall be unique.
- Customers may cancel orders only before shipment.
- Refunds shall be processed after return approval.
- Only authorized administrators can modify inventory data.

---

7. Error Handling Requirements

The system shall display appropriate error messages for:

- Invalid login credentials.
- Payment failures.
- Out-of-stock products.
- Invalid shipping address.
- Network connectivity issues.
- Server errors.

---

8. Security Requirements

- User authentication and authorization.
- Encrypted password storage.
- Secure payment processing.
- Role-based access control.
- Protection of customer data.
- Secure session management.

---

9. Performance Requirements

- System response time should be less than 3 seconds.
- Support concurrent users without performance degradation.
- Real-time inventory updates.
- High system availability.
- Fast order processing during peak traffic.

---

10. Functional Requirement Traceability

Business Requirement	Functional Requirement
Automate order processing	FR-007 Order Creation
Real-time inventory validation	FR-008 Inventory Management
Secure payment processing	FR-006 Payment Processing
Shipment tracking	FR-010 Shipment Tracking
Returns management	FR-011 Returns and Refunds
Business reporting	FR-013 Admin Dashboard

---

11. Expected Outcome

The Functional Requirements defined in this document provide a clear specification for developing an integrated Order Management System that fulfills business needs while delivering a secure, efficient, and user-friendly experience.

---

12. Conclusion

The Functional Requirements Document translates business needs into detailed system functionalities that guide development and testing activities. By clearly defining system behavior, validation rules, and module-level requirements, this document ensures that the final solution aligns with business objectives and stakeholder expectations.

User Stories

1. Introduction

User Stories describe the functional requirements of the E-Commerce Order Management System from the perspective of end users. They help the development team understand user needs and deliver business value through iterative Agile development.

---

2. Purpose

The purpose of User Stories is to capture business requirements in a simple, user-focused format that supports Agile planning, sprint execution, and feature development.

---

3. User Story Format

The following format is used throughout this document:

As a [User Role], I want [Functionality], so that [Business Value].

---

4. Customer User Stories

Story ID	User Story	Priority
US-001	As a customer, I want to register an account so that I can place online orders.	High
US-002	As a customer, I want to log in securely so that I can access my account.	High
US-003	As a customer, I want to search for products so that I can easily find items I need.	High
US-004	As a customer, I want to filter products by category and price so that I can make better purchasing decisions.	Medium
US-005	As a customer, I want to add products to my shopping cart so that I can purchase multiple items together.	High
US-006	As a customer, I want to update quantities in my cart so that I can modify my order before checkout.	Medium
US-007	As a customer, I want to complete secure online payments so that my order can be processed safely.	High
US-008	As a customer, I want to receive an order confirmation so that I know my purchase was successful.	High

Story ID	User Story	Priority
US-009	As a customer, I want to track my order in real time so that I know its delivery status.	High
US-010	As a customer, I want to cancel my order before shipment so that I can change my purchase decision.	Medium
US-011	As a customer, I want to request a return or refund so that I can resolve issues with delivered products.	Medium

5. Administrator User Stories

Story ID	User Story	Priority
US-012	As an administrator, I want to manage product information so that customers always see accurate details.	High
US-013	As an administrator, I want to manage inventory so that stock availability remains accurate.	High
US-014	As an administrator, I want to monitor all orders so that I can oversee business operations.	High
US-015	As an administrator, I want to view business reports so that I can make informed decisions.	High
US-016	As an administrator, I want to manage user accounts so that platform security is maintained.	Medium

6. Warehouse User Stories

Story ID	User Story	Priority
US-017	As a warehouse executive, I want to receive new order notifications so that I can begin fulfillment immediately.	High
US-018	As a warehouse executive, I want to update packing status so that shipment progress is visible to all stakeholders.	Medium
US-019	As a warehouse executive, I want to mark orders as dispatched so that customers receive shipment updates.	High

7. Finance User Stories

Story ID	User Story	Priority
US-020	As a finance executive, I want to verify successful payments so that orders are processed correctly.	High
US-021	As a finance executive, I want to process refunds efficiently so that customer issues are resolved promptly.	Medium

---

8. Customer Support User Stories

Story ID	User Story	Priority
US-022	As a customer support executive, I want to access customer order history so that I can resolve queries quickly.	High
US-023	As a customer support executive, I want to track return requests so that I can provide accurate updates to customers.	Medium

---

9. Delivery Partner User Stories

Story ID	User Story	Priority
US-024	As a delivery partner, I want to receive delivery assignments automatically so that deliveries can be completed efficiently.	High
US-025	As a delivery partner, I want to update delivery status so that customers receive real-time notifications.	High

---

10. Business Value

The implementation of these user stories will:

- Improve customer experience.
- Automate business operations.
- Reduce manual effort.
- Increase order processing efficiency.
- Improve inventory accuracy.
- Enable real-time order visibility.
- Enhance operational productivity.
- Support business growth.

---

11. Story Prioritization

The user stories have been prioritized based on business value and implementation urgency.

- **High Priority:** Essential for the Minimum Viable Product (MVP).
- **Medium Priority:** Important for improving user experience and operational efficiency.
- **Low Priority:** Future enhancements and optional features.

---

12. Conclusion

The User Stories documented in this section provide a clear understanding of user expectations and business needs from multiple stakeholder perspectives. They serve as the foundation for Agile sprint planning, development, testing, and acceptance, ensuring that the E-Commerce Order Management System delivers maximum value to both the business and its users.

Acceptance Criteria

1. Introduction

Acceptance Criteria define the conditions that must be satisfied for a user story or functional requirement to be accepted by the business stakeholders. These criteria ensure that the developed functionality meets the expected business needs and quality standards.

---

2. Purpose

The purpose of Acceptance Criteria is to establish clear and measurable conditions for feature completion, reduce ambiguity during development, and provide a basis for testing and user acceptance.

---

3. Acceptance Criteria Format

Acceptance Criteria are documented using the **Given – When – Then** format.

- **Given:** The initial condition or system state.
- **When:** The action performed by the user.
- **Then:** The expected outcome.

---

4. User Registration (US-001)

Acceptance Criteria

- **Given** a new user is on the registration page,
- **When** valid registration details are entered,
- **Then** the system shall create a new user account successfully.
- **Given** an existing email address is entered,
- **When** the user submits the registration form,
- **Then** the system shall display an appropriate validation message.

---

5. User Login (US-002)

Acceptance Criteria

- **Given** a registered user,
- **When** valid login credentials are entered,

- **Then** the user shall be redirected to the dashboard.

- **Given** invalid credentials,

- **When** the user attempts to log in,
- **Then** the system shall display an error message without granting access.

---

6. Product Search (US-003)

Acceptance Criteria

- **Given** products exist in the catalog,
- **When** the customer searches using a keyword,

- **Then** matching products shall be displayed.

- **Given** no matching products exist,

- **When** the search is performed,
- **Then** the system shall display a "No Products Found" message.

---

7. Shopping Cart (US-005)

Acceptance Criteria

- **Given** a customer selects a product,
- **When** the "Add to Cart" button is clicked,

- **Then** the product shall be added to the shopping cart.

- **Given** products are in the cart,

- **When** quantities are updated,
- **Then** the cart total shall be recalculated automatically.

---

8. Checkout Process

Acceptance Criteria

- **Given** products are available in the shopping cart,
- **When** the customer proceeds to checkout,

- **Then** the system shall display the order summary and shipping details page.

- **Given** mandatory information is missing,

- **When** checkout is attempted,

• **Then** the system shall display validation messages.

---

9. Payment Processing

Acceptance Criteria

• **Given** a valid order,
• **When** payment is completed successfully,

• **Then** the system shall confirm the payment and create the order.

• **Given** payment fails,

• **When** the transaction is unsuccessful,
• **Then** the system shall notify the customer and prevent order creation.

---

10. Order Creation

Acceptance Criteria

• **Given** successful payment,
• **When** the transaction is completed,

• **Then** the system shall generate a unique Order ID.

• **Given** an order is created,

• **When** confirmation is generated,
• **Then** the customer shall receive an order confirmation notification.

---

11. Inventory Management

Acceptance Criteria

• **Given** an order is confirmed,
• **When** inventory is updated,

• **Then** the available stock quantity shall decrease automatically.

• **Given** inventory is unavailable,

• **When** checkout is attempted,
• **Then** the customer shall be informed that the product is out of stock.

---

12. Shipment Tracking

Acceptance Criteria

- **Given** an order has been dispatched,
- **When** the customer views order details,
- **Then** the latest shipment status shall be displayed.
- **Given** delivery status changes,
- **When** tracking information is updated,
- **Then** the customer shall receive a notification.

---

13. Order Cancellation

Acceptance Criteria

- **Given** an order has not been shipped,
- **When** the customer requests cancellation,
- **Then** the system shall cancel the order successfully.
- **Given** an order has already been shipped,
- **When** cancellation is requested,
- **Then** the system shall prevent cancellation and display an appropriate message.

---

14. Returns and Refunds

Acceptance Criteria

- **Given** an eligible delivered order,
- **When** the customer submits a return request,
- **Then** the system shall create a return request successfully.
- **Given** the returned product is approved,
- **When** the refund process begins,
- **Then** the refund status shall be updated and communicated to the customer.

---

15. Admin Dashboard

Acceptance Criteria

- **Given** an administrator logs into the system,

- **When** the dashboard is accessed,
- **Then** key business metrics such as total orders, revenue, pending orders, and inventory status shall be displayed.

---

16. General Acceptance Criteria

- All mandatory fields shall be validated.
- Error messages shall be clear and user-friendly.
- Sensitive data shall be protected.
- Only authorized users shall access restricted functions.
- The system shall maintain data consistency across all modules.
- The application shall support responsive access across desktop and mobile devices.

---

17. Exit Criteria

The feature will be considered complete when:

- All acceptance criteria are successfully met.
- Functional testing is completed successfully.
- No critical defects remain open.
- Business stakeholders approve the functionality.
- User Acceptance Testing (UAT) is successfully completed.

---

18. Conclusion

The Acceptance Criteria provide measurable conditions for validating the successful implementation of business requirements and user stories. They serve as a reference for development, testing, and stakeholder approval, ensuring that the E-Commerce Order Management System meets the expected functional and business objectives.

Process Diagrams

1. Introduction

Process Diagrams provide a visual representation of the business workflows and system interactions within the E-Commerce Order Management System. They help stakeholders understand the sequence of activities, user interactions, decision points, and information flow throughout the order lifecycle.

---

2. Purpose

The purpose of Process Diagrams is to visualize the existing and proposed business processes, improve communication among stakeholders, and provide a clear understanding of how the system operates.

---

3. Context Diagram

Description

The Context Diagram illustrates the E-Commerce Order Management System as a single process and shows its interaction with external entities such as customers, payment gateways, warehouse teams, delivery partners, and administrators.

External Entities:

- Customer
- Payment Gateway
- Warehouse Team
- Delivery Partner
- Administrator

System:

- E-Commerce Order Management System

Data Flow:

- Customer places order
- Payment Gateway processes payment
- Warehouse receives fulfillment request
- Delivery Partner updates shipment status
- Administrator manages system operations

Context Diagram

```
                    +------------------+
                    |     Customer     |
                    +------------------+
```

```
                     |
           Places Order / Track Order
                     |
                     ▼
     +----------------------------------------+
     |    E-Commerce Order Management System  |
     +----------------------------------------+
       ▲            ▲            ▲           ▲
       |            |            |           |
Payment    |   Fulfillment Request   | Shipment  | Reports &
Status     |                         | Status    | Management
           |                         | Updates   |
+----------------+   +----------------+   +----------------+
+----------------+
| Payment        |   | Warehouse      |   | Delivery       |   | Administrator
|
| Gateway        |   | Team           |   | Partner        |   |
|
+----------------+   +----------------+   +----------------+
+----------------+
```

---

4. Use Case Diagram

Description

The Use Case Diagram identifies the system users (actors) and the functionalities they can perform within the Order Management System.

Actors:

- Customer
- Administrator
- Warehouse Executive
- Finance Executive
- Delivery Partner

Use Cases:

- Register Account
- Login
- Search Products
- Add to Cart
- Checkout
- Make Payment
- Track Order
- Cancel Order
- Request Return
- Manage Inventory
- Generate Reports

Use Case Diagram

```
                    +-------------------------------------------------+
                    |         Order Management System                 |
                    |                                                 |
                    |  (Register Account)                             |
                    |  (Login)                                        |
                    |  (Search Products)                              |
                    |  (Add to Cart)                                  |
                    |  (Checkout)                                     |
                    |  (Make Payment)                                 |
                    |  (Track Order)                                  |
                    |  (Cancel Order)                                 |
                    |  (Request Return)                               |
                    |  (Manage Inventory)                             |
                    |  (Generate Reports)                             |
                    +-------------------------------------------------+

Customer --------------------> Register Account
Customer --------------------> Login
Customer --------------------> Search Products
Customer --------------------> Add to Cart
Customer --------------------> Checkout
Customer --------------------> Make Payment
Customer --------------------> Track Order
Customer --------------------> Cancel Order
Customer --------------------> Request Return

Administrator ---------------> Manage Inventory
Administrator ---------------> Generate Reports

Warehouse Executive ---------> Manage Inventory

Finance Executive -----------> Make Payment
Finance Executive -----------> Generate Reports

Delivery Partner ------------> Track Order
```

---

5. Activity Diagram

Description

The Activity Diagram represents the sequence of activities performed during the order placement process.

Main Activities:

- Customer Login

- Browse Products
- Add to Cart
- Checkout
- Payment
- Order Creation
- Inventory Update
- Warehouse Processing
- Shipment Dispatch
- Order Delivery

Activity Diagram

```
                       +-------------------+
                       |       Start       |
                       +-------------------+
                                |
                                v
                    +-----------------------+
                    |   Customer Login      |
                    +-----------------------+
                                |
                                v
                    +-----------------------+
                    |  Browse Products      |
                    +-----------------------+
                                |
                                v
                    +-----------------------+
                    |   Add Items to Cart   |
                    +-----------------------+
                                |
                                v
                    +-----------------------+
                    |      Checkout         |
                    +-----------------------+
                                |
                                v
                    +-----------------------+
                    |   Make Payment        |
                    +-----------------------+
                                |
                       +--------+--------+
                       | Payment Success?|
                       +--------+--------+
                          Yes   |   No
                                |
            +-------------------+-------------------+
            |                                       |
            v                                       v
+-------------------------+            +-------------------------+
|     Create Order        |            | Payment Failed          |
```

```
   +-------------------------+          | Retry/Cancel Payment   |
               |                        +------------------------+
               |                                      |
               +--------------------+-----------------+
                                    |
                                    v
                       +----------------------+
                       |   Update Inventory   |
                       +----------------------+
                                   |
                                   v
                       +----------------------+
                       | Warehouse Processing |
                       +----------------------+
                                   |
                                   v
                       +----------------------+
                       |  Shipment Dispatch   |
                       +----------------------+
                                   |
                                   v
                       +----------------------+
                       |   Order Delivered    |
                       +----------------------+
                                   |
                                   v
                          +------------------+
                          |       End        |
                          +------------------+
```

---

6. BPMN Process Diagram

Description

The BPMN (Business Process Model and Notation) Diagram illustrates the end-to-end business workflow using standard BPMN symbols and decision gateways.

Major Process Steps:

- Customer places order
- Payment validation
- Inventory validation
- Warehouse fulfillment
- Shipment assignment
- Delivery confirmation
- Return and refund process

BPMN Process Diagram

```
┌───────────────┐
│     Start     │
└───────┬───────┘
        │
        ▼
┌─────────────────────────┐
│ Customer Places Order   │
└──────────┬──────────────┘
           │
           ▼
┌─────────────────────────┐
│ Validate Payment        │
└──────────┬──────────────┘
           │
           ▼
      ┌───────────┐
      │ Payment ? │
      └─────┬─────┘
       Yes  │  No
            │
            ▼
     ┌───────────────┐
     │ Payment Failed│
     └──────┬────────┘
            │
            ▼
        ┌─────────┐
        │   End   │
        └─────────┘

 (Yes Path)

            │
            ▼
┌─────────────────────────┐
│ Validate Inventory      │
└──────────┬──────────────┘
           │
           ▼
      ┌─────────────────┐
      │ Stock Available?│
      └─────┬───────────┘
      Yes   │   No
            │
            ▼
   ┌───────────────────────┐
   │ Notify Out of Stock   │
```

```
    └───────────────────────┘
              │
              │
              ▼
        ┌──────────┐
        │   End    │
        └──────────┘

(Yes Path)

             │
             ▼
┌──────────────────────────┐
│ Warehouse Fulfillment    │
└──────────────────────────┘
           │
           │
           ▼
┌──────────────────────────┐
│ Assign Shipment          │
└──────────────────────────┘
           │
           │
           ▼
┌──────────────────────────┐
│ Deliver Order            │
└──────────────────────────┘
           │
           │
           ▼
┌──────────────────────────┐
│ Delivery Confirmation    │
└──────────────────────────┘
           │
           │
           ▼
      ┌──────────────────┐
      │ Return Request?  │
      └──────────────────┘
         Yes │ No
             │
             ▼
       ┌──────────┐
       │   End    │
       └──────────┘

(Yes Path)

             │
             ▼
┌──────────────────────────┐
│ Process Return           │
└──────────────────────────┘
           │
           │
           ▼
```

```
┌─────────────────────────────┐
│ Refund Customer             │
└─────────────────────────────┘
            │
            │
            ▼
     ┌──────────┐
     │   End    │
     └──────────┘
```

---

7. Swimlane Diagram

Description

The Swimlane Diagram shows how responsibilities are distributed across different stakeholders during the order management process.

Swimlanes:

- Customer
- Order Management System
- Payment Gateway
- Warehouse Team
- Delivery Partner

The diagram demonstrates how work moves between departments and external systems.

Swimlane Diagram

```
+-----------------+-------------------------+------------------
+------------------+------------------+
|    Customer     | Order Management System | Payment Gateway  |  Warehouse
Team   | Delivery Partner |
+-----------------+-------------------------+------------------
+------------------+------------------+

      Start
        |
        ▼
+-----------------+
| Browse Products|
+-----------------+
        |
        ▼
+-----------------+
| Add to Cart     |
+-----------------+
        |
        ▼
```

```
+----------------+
| Checkout Order |
+----------------+
        |
        ├──────────────────────▶
                         +---------------------+
                         | Create Order        |
                         +---------------------+
                                   |
                                   ├──────────────────▶
                                            +-----------------+
                                            | Process Payment |
                                            +-----------------+
                                                    |
                              ◀─────────────────────┤
                              Payment Success
                                   |
                                   ▼
                         +---------------------+
                         | Confirm Order       |
                         +---------------------+
                                   |
                                   ├───────────────────────▶

+---------------------+
                                            | Pick & Pack Order
|

+---------------------+
                                                    |
                                                    ▼

+---------------------+
                                            | Dispatch Order
|

+---------------------+
                                                    |
                               ◀────────────────────┤
                               Dispatch Details
                                   |

├────────────────────────────────▶

+-----------------+
                                                            |
Deliver Order    |

+-----------------+

|
```

```
▼

+------------------+
                                                                            |
Update Status    |

+------------------+

|

◄───────────────────────────────────────────┤
                                    Delivery Confirmation
                                          |
                                          ▼
                              +----------------------+
                              | Update Order Status  |
                              +----------------------+
                                          |
                                          ▼
+----------------+
| Track Order    |
+----------------+
        |
        ▼
+----------------+
| End            |
+----------------+

+----------------+------------------------+------------------
+------------------+------------------+
```

8. Order Lifecycle Diagram

Description

The Order Lifecycle Diagram represents the different states through which an order passes from creation to completion.

Order Status Flow:

Created

↓

Payment Successful

↓

Confirmed

↓

Packed

↓

Shipped

↓

Out for Delivery

↓

Delivered

↓

Completed

Alternative Flow:

Created

↓

Cancelled

or

Delivered

↓

Return Requested

↓

Refund Processed

Order Lifecycle Diagram

```
            ┌──────────────────┐
            │     Created      │
            └──────────────────┘
                     │
```

```
                  ▼
       ┌──────────────────────┐
       │  Payment Successful  │
       └──────────────────────┘
                  │
                  ▼
           ┌──────────────────┐
           │    Confirmed     │
           └──────────────────┘
                  │
                  ▼
           ┌──────────────────┐
           │      Packed      │
           └──────────────────┘
                  │
                  ▼
           ┌──────────────────┐
           │     Shipped      │
           └──────────────────┘
                  │
                  ▼
           ┌──────────────────┐
           │ Out for Delivery │
           └──────────────────┘
                  │
                  ▼
           ┌──────────────────┐
           │    Delivered     │
           └──────────────────┘
                  │
                  ▼
           ┌──────────────────┐
           │    Completed     │
           └──────────────────┘
```

Alternative Flow 1:

```
           ┌──────────────────┐
           │     Created      │
           └──────────────────┘
                  │
                  ▼
           ┌──────────────────┐
           │    Cancelled     │
           └──────────────────┘
```

Alternative Flow 2:

```
┌──────────────────┐
│     Delivered    │
└──────────────────┘
         │
         ▼
┌──────────────────┐
│ Return Requested │
└──────────────────┘
         │
         ▼
┌──────────────────┐
│ Refund Processed │
└──────────────────┘
```

9. Process Improvement Comparison

AS-IS Process	TO-BE Process
Manual order processing	Automated order processing
Manual inventory validation	Real-time inventory validation
Manual shipment assignment	Automatic shipment assignment
Delayed status updates	Real-time order tracking
Disconnected systems	Integrated centralized platform
Manual reporting	Automated dashboards and analytics

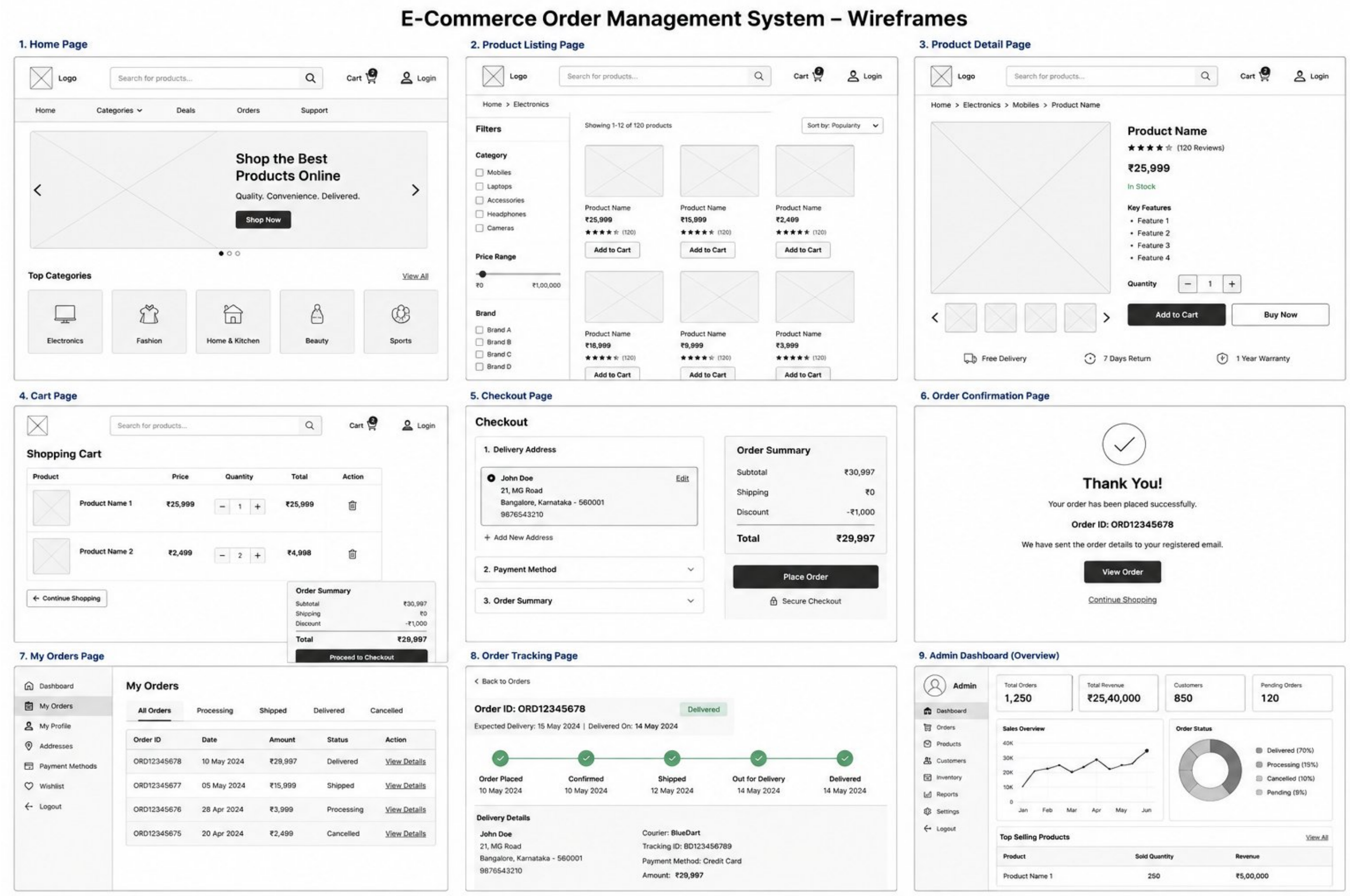
10. Key Benefits of Process Diagrams

- Improves understanding of business workflows.
- Identifies bottlenecks and inefficiencies.
- Supports requirement validation.
- Enhances stakeholder communication.
- Assists developers in understanding business logic.
- Simplifies testing and process verification.
- Supports future process optimization.

11. Conclusion

The Process Diagrams provide a comprehensive visual representation of the E-Commerce Order Management System and its business workflows. They improve communication between business and technical teams, facilitate requirement validation, and support the successful implementation of an efficient and scalable order management solution.

Wireframes


E-Commerce Order Management System – Wireframes
1. Home Page
2. Product Listing Page
3. Product Detail Page
4. Cart Page
5. Checkout Page
6. Order Confirmation Page
7. My Orders Page
8. Order Tracking Page
9. Admin Dashboard (Overview)

Test Scenarios

1. Introduction

Test Scenarios define the high-level conditions and business flows that need to be validated to ensure the E-Commerce Order Management System functions according to the specified business and functional requirements. These scenarios support Quality Assurance (QA) activities and User Acceptance Testing (UAT).

---

2. Purpose

The purpose of Test Scenarios is to:

- Validate business requirements.
- Verify end-to-end business processes.
- Ensure system functionality meets user expectations.
- Identify defects before production deployment.
- Support User Acceptance Testing (UAT).

---

3. Scope

The Test Scenarios cover the major modules of the E-Commerce Order Management System, including:

- User Management
- Product Catalog
- Shopping Cart
- Checkout
- Payment Processing
- Order Management
- Inventory Management
- Warehouse Processing
- Shipment Tracking
- Returns and Refunds
- Reporting Dashboard

---

4. Test Scenarios

TS ID	Test Scenario	Expected Result	Priority
TS-001	Verify new user registration	User account is created successfully	High
TS-002	Verify user login with valid credentials	User is logged into the system	High

TS ID	Test Scenario	Expected Result	Priority
TS-003	Verify login with invalid credentials	Appropriate error message is displayed	High
TS-004	Verify product search functionality	Matching products are displayed	High
TS-005	Verify product filtering	Filtered products are displayed correctly	Medium
TS-006	Verify adding products to cart	Selected products are added to the cart	High
TS-007	Verify updating cart quantity	Cart quantity and total are updated	Medium
TS-008	Verify removing products from cart	Product is removed successfully	Medium
TS-009	Verify checkout process	Order summary page is displayed	High
TS-010	Verify successful payment	Payment is processed and order is created	High
TS-011	Verify failed payment transaction	Payment failure message is displayed	High
TS-012	Verify order confirmation	Customer receives order confirmation	High
TS-013	Verify inventory update after order	Inventory quantity is reduced correctly	High
TS-014	Verify warehouse order processing	Order status changes to Packed	Medium
TS-015	Verify shipment dispatch	Order status changes to Shipped	Medium
TS-016	Verify order tracking	Latest shipment status is displayed	High
TS-017	Verify order cancellation before shipment	Order is cancelled successfully	High
TS-018	Verify cancellation after shipment	Cancellation request is rejected	Medium
TS-019	Verify return request submission	Return request is created successfully	Medium
TS-020	Verify refund processing	Refund status is updated successfully	Medium
TS-021	Verify administrator dashboard	Business metrics are displayed correctly	High
TS-022	Verify report generation	Reports are generated successfully	Medium

5. User Acceptance Test (UAT) Scenarios

UAT ID	Business Scenario	Expected Outcome
UAT-001	Customer completes end-to-end purchase	Order is successfully delivered

UAT ID	Business Scenario	Expected Outcome
UAT-002	Customer tracks an active order	Real-time order status is visible
UAT-003	Customer cancels an eligible order	Order is cancelled successfully
UAT-004	Customer requests a return	Return request is accepted
UAT-005	Administrator manages inventory	Inventory updates correctly
UAT-006	Administrator generates reports	Reports display accurate business data

6. Entry Criteria

Testing can begin when:

- Business Requirements are approved.
- Functional Requirements are finalized.
- Development is completed.
- Test environment is available.
- Test data has been prepared.

7. Exit Criteria

Testing is considered complete when:

- All planned test scenarios have been executed.
- All critical and high-priority defects are resolved.
- Acceptance criteria are satisfied.
- User Acceptance Testing (UAT) is approved by stakeholders.
- Business stakeholders provide sign-off for production deployment.

8. Defect Severity Classification

Severity	Description
Critical	System failure or major business impact
High	Core functionality not working
Medium	Functional issue with workaround available
Low	Minor UI or cosmetic issue

9. Test Execution Summary

Category	Total
Total Test Scenarios	22
High Priority	11
Medium Priority	11
Low Priority	0
UAT Scenarios	6

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10. Benefits of Test Scenarios

- Ensures complete validation of business requirements.
- Improves software quality and reliability.
- Identifies defects early in the testing cycle.
- Supports requirement traceability.
- Reduces production risks.
- Increases stakeholder confidence before deployment.

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11. Conclusion

The Test Scenarios document provides a structured approach for validating the functionality and business processes of the E-Commerce Order Management System. By covering key business flows and user interactions, it ensures that the solution meets business objectives, supports successful User Acceptance Testing, and delivers a reliable and high-quality system to end users.

Business Benefits

1. Introduction

The implementation of the E-Commerce Order Management System (OMS) is expected to deliver significant business value by automating order processing, improving operational efficiency, enhancing customer satisfaction, and enabling better decision-making through real-time data and reporting. This document outlines the key business benefits that the organization can achieve after implementing the proposed solution.

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2. Purpose

The purpose of this document is to:

- Identify the expected business value of the project.
- Demonstrate how the proposed solution aligns with business objectives.
- Measure improvements over the existing process.
- Support project justification and stakeholder approval.
- Highlight both qualitative and quantitative benefits.

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3. Business Objectives Achieved

The proposed solution helps the organization achieve the following objectives:

- Automate the complete order lifecycle.
- Improve order processing efficiency.
- Reduce manual intervention and operational errors.
- Enhance inventory visibility and control.
- Provide real-time order tracking.
- Improve customer experience and satisfaction.
- Enable data-driven business decisions.
- Support future business growth and scalability.

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4. Operational Benefits

Current Process (AS-IS)	Future Process (TO-BE)
Manual order processing	Automated order processing
Manual inventory updates	Real-time inventory synchronization
Separate business systems	Integrated centralized platform
Delayed order tracking	Real-time order tracking

Current Process (AS-IS)	Future Process (TO-BE)
Manual reporting	Automated dashboards and reports
High dependency on manual work	Increased automation and efficiency

5. Business Benefits

Improved Operational Efficiency

- Reduces manual effort through workflow automation.
- Standardizes business processes across departments.
- Improves coordination between warehouse, finance, and delivery teams.
- Reduces processing delays and bottlenecks.

Faster Order Processing

- Enables quicker order confirmation and fulfillment.
- Reduces processing time from order placement to shipment.
- Improves warehouse productivity through automated task allocation.

Better Inventory Management

- Provides real-time inventory visibility.
- Reduces stock discrepancies.
- Prevents overselling and stock shortages.
- Supports efficient inventory planning.

Enhanced Customer Experience

- Simplifies the online purchasing process.
- Provides instant order confirmation.
- Enables real-time shipment tracking.
- Improves transparency throughout the order lifecycle.
- Supports easy order cancellation and return requests.

Improved Data Accuracy

- Eliminates duplicate data entry.
- Reduces manual errors.
- Ensures consistent information across all business functions.
- Improves overall data quality.

Better Reporting and Analytics

- Provides centralized business reports.
- Enables real-time performance monitoring.
- Supports faster management decision-making.
- Improves visibility into sales, inventory, and order trends.

Increased Productivity

- Allows employees to focus on higher-value activities.
- Reduces repetitive administrative tasks.
- Improves collaboration across departments.
- Optimizes resource utilization.

Cost Savings

- Reduces operational costs associated with manual processing.
- Minimizes errors and rework.
- Lowers customer support costs through self-service tracking.
- Improves overall process efficiency.

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6. Customer Benefits

Customers benefit from the proposed system through:

- Faster order confirmation.
- Secure online payment processing.
- Accurate product availability information.
- Real-time order tracking.
- Simplified return and refund process.
- Improved shopping experience.
- Greater transparency and communication.

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7. Organizational Benefits

The organization benefits through:

- Increased operational efficiency.
- Higher customer satisfaction.
- Improved employee productivity.
- Better inventory control.
- Enhanced business intelligence.
- Increased revenue opportunities.
- Stronger competitive advantage.
- Greater scalability for future expansion.

8. Key Performance Indicators (KPIs)

KPI	Expected Improvement
Order Processing Time	Reduced by 50%
Manual Processing Effort	Reduced by 70%
Inventory Accuracy	Increased to 98%+
Order Fulfillment Speed	Improved by 40%
Customer Satisfaction	Increased significantly
Reporting Time	Reduced from hours to minutes
Operational Efficiency	Improved across departments

9. Return on Investment (ROI)

The implementation of the Order Management System is expected to generate long-term value by:

- Reducing operational costs.
- Increasing employee productivity.
- Improving order fulfillment performance.
- Enhancing customer retention.
- Supporting business growth.
- Delivering measurable process improvements.

Although the initial investment includes system development, implementation, and training, the long-term operational savings and business benefits are expected to outweigh these costs.

10. Strategic Impact

The proposed solution supports the organization's long-term strategy by:

- Enabling digital transformation.
- Improving business agility.
- Supporting scalable growth.
- Enhancing customer-centric operations.
- Promoting data-driven decision-making.
- Increasing competitiveness in the e-commerce market.

11. Summary of Benefits

Benefit Area	Expected Outcome
Order Management	Faster and automated processing
Inventory Management	Accurate real-time stock visibility
Customer Experience	Improved satisfaction and transparency
Business Operations	Increased efficiency and productivity
Reporting	Real-time analytics and dashboards
Cost Management	Reduced operational expenses
Decision Making	Better insights through centralized data
Business Growth	Scalable platform for future expansion

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12. Conclusion

The E-Commerce Order Management System delivers substantial business value by streamlining operations, improving process efficiency, enhancing customer experience, and enabling informed decision-making through integrated data and reporting capabilities. The proposed solution not only addresses current business challenges but also establishes a scalable and future-ready platform that supports sustainable organizational growth and competitive advantage.

Conclusion

1. Introduction

The E-Commerce Order Management System case study demonstrates a comprehensive Business Analysis approach for designing an efficient, scalable, and customer-centric solution to manage the complete order lifecycle. The project showcases how structured analysis and documentation can help organizations improve operational efficiency, streamline business processes, and achieve strategic objectives.

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2. Project Summary

The proposed solution integrates customer order placement, payment processing, inventory validation, warehouse fulfillment, shipment tracking, returns management, and reporting into a centralized platform. By replacing fragmented and manual processes with an automated system, the organization can improve productivity, reduce errors, and deliver a seamless customer experience.

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3. Key Deliverables

The project includes the following Business Analysis artifacts:

- Project Overview
- Business Case
- Stakeholder Analysis
- AS-IS Process Analysis
- TO-BE Process Analysis
- Requirements Elicitation
- Business Requirements Document (BRD)
- Functional Requirements Document (FRD)
- User Stories
- Acceptance Criteria
- Process Diagrams
- Wireframes
- Test Scenarios
- Business Benefits

These deliverables collectively provide a structured framework for understanding business needs, defining solution requirements, and supporting successful project implementation.

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4. Business Value Delivered

The proposed solution is expected to deliver significant business value through:

- Automation of the end-to-end order management process.

• Faster order processing and fulfillment.
• Improved inventory accuracy and real-time visibility.
• Reduction in manual effort and operational errors.
• Enhanced customer satisfaction through transparent order tracking.
• Better reporting and data-driven decision-making.
• Increased operational efficiency across departments.

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5. Business Analyst Contribution

This case study demonstrates the critical role of a Business Analyst in the successful delivery of business solutions by:

• Understanding business problems and objectives.
• Identifying and analyzing stakeholder requirements.
• Conducting process analysis and gap identification.
• Documenting business and functional requirements.
• Modeling business processes using standard diagrams.
• Defining user stories and acceptance criteria.
• Supporting requirement traceability and testing activities.
• Recommending process improvements aligned with organizational goals.

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6. Final Outcome

The proposed E-Commerce Order Management System provides a scalable and integrated platform that addresses existing business challenges while supporting future growth. By improving collaboration between customers, finance, warehouse operations, delivery partners, and administrators, the system enables efficient order management and enhances overall business performance.

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7. Project Success Factors

The success of the proposed solution depends on:

• Clearly defined business requirements.
• Effective stakeholder collaboration.
• Well-documented functional specifications.
• Standardized business processes.
• Comprehensive testing and validation.
• Continuous process improvement and monitoring.

These factors contribute to the successful implementation and long-term sustainability of the solution.

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8. Lessons Learned

The development of this case study highlights the importance of:

- Early stakeholder engagement.
- Accurate requirement gathering and analysis.
- Clear documentation and traceability.
- Process modeling for better business understanding.
- Alignment between business objectives and technical solutions.
- Continuous communication throughout the project lifecycle.

These practices help minimize project risks and improve solution quality.

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9. Closing Statement

This case study represents a complete Business Analysis lifecycle, from identifying business problems to designing an optimized future-state solution. It demonstrates the practical application of Business Analysis methodologies, documentation standards, and process modeling techniques required for successful project delivery.

The E-Commerce Order Management System serves as a comprehensive example of how Business Analysis contributes to digital transformation by bridging the gap between business needs and technology solutions while delivering measurable business value and enhanced customer satisfaction.

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10. Conclusion

The E-Commerce Order Management System case study showcases a structured and professional Business Analysis approach that supports effective decision-making, process optimization, and successful solution implementation. Through detailed analysis, comprehensive documentation, and standardized modeling techniques, the project demonstrates the essential skills and responsibilities of a Business Analyst in delivering business-focused technology solutions. The proposed system establishes a strong foundation for operational excellence, business scalability, and long-term organizational success.